

## Manuel Francesco Aprile

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| <b>Contacts</b>               | Date and place of birth:                                                                | September 23, 1991, Mascalucia (CT), Italy                                                                                                              |
|                               | Nationality:                                                                            | Italian                                                                                                                                                 |
|                               | Email:                                                                                  | manuel.aprile@unipd.it                                                                                                                                  |
|                               | Phone:                                                                                  | +39 3384327987                                                                                                                                          |
|                               | Web:                                                                                    | <a href="https://manuel-aprile.github.io/my_website/">https://manuel-aprile.github.io/my_website/</a>                                                   |
|                               | Office:                                                                                 | 4BC7, Department of Mathematics<br>University of Padua<br>Via Trieste, 63<br>Padua, Italy                                                               |
|                               | ORCID:                                                                                  | 0000-0002-6805-6903                                                                                                                                     |
|                               |                                                                                         |                                                                                                                                                         |
| <b>Research<br/>Interests</b> | Combinatorial optimization, polyhedra, extended formulations, graph and matroid theory. |                                                                                                                                                         |
| <b>Employment</b>             | Since Nov. 2025:                                                                        | Associate Professor<br>University of Padua, Italy                                                                                                       |
|                               | Since Dec. 2022:                                                                        | Tenure-track Researcher (RTD-B)<br>University of Padua, Italy                                                                                           |
|                               | Sept.–Dec. 2022:                                                                        | Postdoctoral researcher<br>Department of Mathematics, ULB, Belgium                                                                                      |
|                               | 2020–2022:                                                                              | Research Fellow<br>University of Padua, Italy                                                                                                           |
|                               | 2019–2020:                                                                              | Postdoctoral researcher<br>Department of Mathematics, ULB, Belgium                                                                                      |
|                               | 2014–2018:                                                                              | PhD student (“assistant doctorant”)<br>EPFL, Lausanne, Switzerland                                                                                      |
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| <b>Education</b>              | 2018:                                                                                   | PhD in Discrete Optimization<br>EPFL, Switzerland<br>Advisor: Prof. Friedrich Eisenbrand<br>Co-advisor: Prof. Yuri Faenza                               |
|                               | 2014:                                                                                   | Master of Science in Mathematics and Foundations of Computer Science<br>University of Oxford, UK<br>Advisor: Prof. Colin McDiarmid                      |
|                               | 2013:                                                                                   | Bachelor in Mathematics, <i>summa cum laude</i><br>University of Catania, Italy<br>Advisor: Prof. Giuseppe Nicosia<br>Co-advisor: Prof. Vittorio Romano |
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## Awards and Scholarships

2020: Winner of the Lorenzo Brunetta Prize (2500€) from the Istituto Veneto di Scienze, Lettere ed Arti for my PhD thesis.

2010–13: INdAM Scholarship to support my bachelor studies.

### Mathematics Olympiads:

2010: Bronze medal at the Italian National Competition, Cesenatico.

2005–09: Selected for the provincial competition, Eastern Sicily.

2004: Selected for the national competition, Bocconi University, Milan.

### Physics Olympiads:

2007–09: Selected for the provincial competition, Eastern Sicily.

## Qualifications

2024: Italian National Scientific Qualification (ASN)

as Associate Professor (Sector 01/A6 – Operations Research).

2020: French National Council of Universities (CNU) Qualification.

*Maître de conférences*, Sections 25 (Mathematics), 26 (Applied Mathematics), 27 (Computer Science).

## Teaching Activities

### Teaching at the University of Padua:

Spring 2021–2025: “Discrete Mathematics” (Bachelor in Mathematics)

Autumn 2023–25: “Fundamentals of Computational Mathematics” (MSc Computational Finance)

Autumn 2025: “Foundations of Mathematics” (Bachelor in Biology)

Spring 2024: “Linear and Non-linear Formulations for Combinatorial Optimization” (PhD course)

### Teaching Assistant at ULB:

Autumn 2019: “Convex and Discrete Geometry” (Master, in French)

### Teaching Assistant at EPFL:

Spring 2015–2018: Discrete Optimization

Autumn 2017: Combinatorial Geometry (Master)

Autumn 2016: Combinatorial Optimization (Master)

Autumn 2015: Linear Algebra (in French)

### Teaching Training Activities:

Feb. 2024: T4L New Faculty and Basic Course, Padua

June 2017: SOTL Workshop, Zurich (invited speaker)

Autumn 2016: Science and Engineering Teaching and Learning (semester course)

May 2015: Instructional Skills Workshop (three-day training)

Oct. 2014: Teaching Toolkit for Doctoral Assistants (one-day training)

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| <b>Supervision of Students</b> | <p>University of Padua: Advisor or co-advisor of about a dozen Bachelor and Master theses defended between 2021 and 2025.</p> <p>EPFL: Supervisor of two Bachelor projects and two Master projects</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Outreach Activities</b>     | Spring 2023: Seminar titled “A Glimpse into Graph Theory” for class 4G, Liceo Matematico “Bruno - Franchetti”, Venice.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Peer Review Activity</b>    | Reviewer for scientific journals and international conferences, including: SIAM Journal on Discrete Mathematics, Discrete & Computational Geometry, European Journal of Combinatorics, Discrete Mathematics & Theoretical Computer Science, Discrete Applied Mathematics, Theoretical Computer Science, Operations Research Letters, IPCO (Integer Programming and Combinatorial Optimization), ISCO, MFCS.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Projects</b>                | Principal Investigator of the SID project “Network Optimization Problems under Extra Constraints”.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Conference Organization</b> | <p>Local organizer of <i>IPCO 2026</i>, Padua.</p> <p>Local organizer of <i>Airo Young 10th Workshop 2026</i>, Padua.</p> <p>Organizer and chair of the session <i>Integer Programming: Structure and Algorithms</i> at ISMP (Montreal, 2024).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Publications</b>            | <p>Conference Proceedings (peer-reviewed):</p> <ul style="list-style-type: none"> <li>• M. Aprile, S. Fiorini, G. Joret, S. Kober, M. Seweryn, L. Yuditsky, S. Weltge, <i>Integer Programs with Nearly Totally Unimodular Matrices: The Cographic Case</i>, SODA 2025.</li> <li>• M. Aprile, <i>Exact Approaches for the Connected Vertex Cover Problem</i>, CTW 2024.</li> <li>• M. Aprile, M. Drescher, S. Fiorini, T. Huynh, <i>A Tight Approximation Algorithm for the Cluster Vertex Deletion Problem</i>, IPCO 2021.</li> <li>• M. Aprile, M. Conforti, Y. Faenza, S. Fiorini, T. Huynh, M. Macchia, <i>Recognizing Cartesian Products of Matrices and Polytopes</i>, Cologne-Twente Workshop 2020.</li> <li>• M. Aprile, Y. Faenza, <i>Extended Formulations from Communication Protocols in Output-efficient Time</i>, IPCO 2019.</li> <li>• M. Aprile, Y. Faenza, S. Fiorini, T. Huynh, M. Macchia, <i>Extension Complexity of Stable Set Polytopes of Bipartite Graphs</i>, WG 2017.</li> <li>• M. Aprile, N. Castro, F. Robledo, P. Romero, <i>Analysis of Node-Resilience Strategies under Natural Disasters</i>, DRCN 2017.</li> <li>• M. Aprile, A. Cevallos, Y. Faenza, <i>On Vertices and Facets of Combinatorial 2-Level Polytopes</i>, ISCO 2016.</li> </ul> |

Journals:

- M. Aprile, G. Averkov, M. Di Summa, C. Hojny, *The Role of Rationality in Integer-Programming Relaxations*, Mathematical Programming B, 2024.
- M. Aprile, M. Conforti, M. Di Summa, *Binary Extended Formulations and Sequential Convexification*, Mathematics of Operations Research, 2024.
- M. Aprile, M. Conforti, Y. Faenza, S. Fiorini, T. Huynh, M. Macchia, *Slack Matrices,  $k$ -Products, and  $k$ -Level Polytopes*, Discrete Applied Mathematics, 2024.
- M. Aprile, M. Drescher, S. Fiorini, T. Huynh, *A Simple  $7/3$ -Approximation Algorithm for Feedback Vertex Set in Tournaments*, Discrete Applied Mathematics, 2023.
- M. Aprile, S. Fiorini, *Regular Matroids Have Polynomial Extension Complexity*, Mathematics of Operations Research, 2022.
- M. Aprile, *Extended Formulations for Matroid Polytopes through Randomized Protocols*, Operations Research Letters, 2022.
- M. Aprile, M. Drescher, S. Fiorini, T. Huynh, *A Tight Approximation Algorithm for the Cluster Vertex Deletion Problem*, Mathematical Programming B, 2022.
- M. Aprile, S. Fiorini, T. Huynh, G. Joret, D. R. Wood, *Smaller Extended Formulations for Spanning Tree Polytopes in Minor-closed Classes and Beyond*, Electronic Journal of Combinatorics, 2021.
- M. Aprile, Y. Faenza, *Extended Formulations from Communication Protocols in Output-efficient Time*, Mathematical Programming B, 2020.
- M. Aprile, A. Cevallos, Y. Faenza, *On 2-level Polytopes Arising in Combinatorial Settings*, SIAM Journal on Discrete Mathematics, 2018.
- M. Aprile, N. Castro, G. Ferreira, J. Piccini, F. Robledo, P. Romero, *Graph Fragmentation Problem: Analysis and Synthesis*, International Transactions in Operational Research, 2018.

Theses:

- M. Aprile, *On Some Problems Related to 2-Level Polytopes*, PhD Thesis, EPFL, 2018.
- M. Aprile, *Constructive Aspects of Lovász Local Lemma and Applications to Graph Colouring*, MSc Thesis, University of Oxford, 2014.
- M. Aprile, *An Algorithm for Constructing Magic Squares*, Bachelor Thesis, University of Catania, 2013.

**Invited and  
Contributed  
Talks**

CTW 2023, Garmisch-Partenkirchen, Germany.  
Cargese Workshop on Combinatorial Optimization 2022, France (invited).  
Separating Integer Points in Polyhedra, 2022, Eindhoven, Netherlands (invited).  
MIP 2022 Mixed Integer Programming Workshop, Rutgers University, USA (invited).  
Cologne-Twente Workshop on Graphs and Combinatorial Optimization 2020.  
Extended Formulations and Symmetries 2019, Rancagua, Chile (invited).  
IPCO 2019, Ann Arbor, USA.  
Cargese Workshop on Combinatorial Optimization (2017–2019), France (invited).  
Aussois Combinatorial Optimization Workshop (2018–2020), France (invited).  
DRCN 2017, Munich, Germany.  
ISCO 2016, Vietri sul Mare, Italy.

**Computer Skills** Programming experience with Python, Matlab, C, Gurobi, L<sup>A</sup>T<sub>E</sub>X.

**Languages** Italian (native), English (advanced), French (intermediate).